London Ambulance Service NHS
NHS Trust

Understand Medical and Surgical Conditions – Component 5

Urinary conditions

Urinary Conditions - Component 5

Objective

Understand urinary conditions in accordance with own scope of practice.

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Acute Urinary Retention (AUR)

Urinary retention is the inability to empty the bladder and can be acute or chronic

Acute urinary retention is a medical emergency

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Acute Urinary Retention (AUR)

Causes

- Prostate enlargement - benign prostate hyperplasia
- Urethral stricture - inflammation/scar tissue from surgery
- Urinary tract stones - blocked opening to urethra
- Nerve problems - vaginal childbirth, spinal cord injury, pelvic injury/trauma
- Aging - Inability to empty bladder completely

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Acute Urinary Retention (AUR)

Signs & Symptoms

- Severe lower abdominal pain
- Unable to pass urine
- Reduced flow
- Signs of infection
- Swelling of the bladder
- Without prompt treatment urine may pass back up into the kidneys, which may damage them

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Urinary Tract Infections (UTI)

A UTI develops when part of the urinary tract becomes infected, usually by bacteria.

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Urinary Tract Infections (UTI)

- UTIs mainly occur in women

- Are far less common in men and for that reason need investigating to find an underlying cause :

  Narrowing of the urethra (a stricture),
  Previous STI,
  Bladder stones,
  Problem with the prostate gland.

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Urinary Tract Infections (UTI)

Lower or upper UTI'S:

- Lower urinary tract (bladder and urethra)
- Upper urinary tract (kidney and ureters)

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Urinary Tract Infections (UTI)

Upper UTIs are potentially more serious than lower UTIs because there is a risk of kidney damage

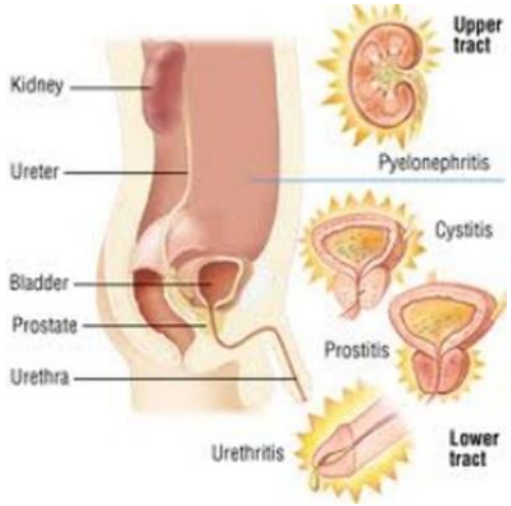
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Urinary Tract Infections (UTI)

Signs & Symptoms

- Pain/burning sensation when urinating
- Cramping pain in lower abdomen or back
- Dysuria – burning sensation
- Increased frequency
- Confusion in the elderly
- Strong smelling urine

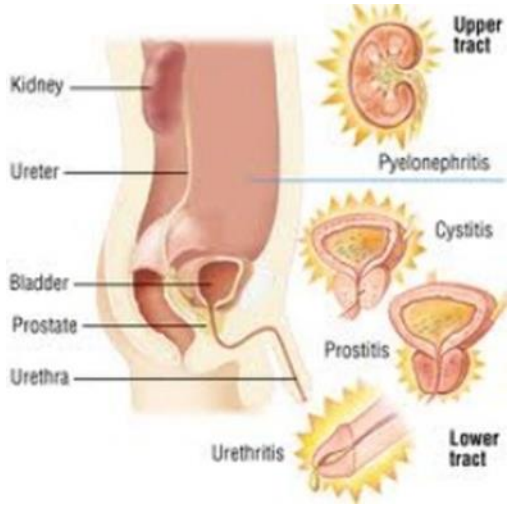


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Urinary Tract Infections (UTI)

- Haematuria - cloudy urine or frank blood
- ? kidney infection: temp >38, rigors, vomiting
- Nausea and vomiting



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Kidney Stones (renal calculi)

- Kidney stones are stone-like lumps that can develop in one or both of the kidneys
- Stone moves down ureters and stretches the wall
- Often with nausea and vomiting



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Kidney Stones (renal calculi)

- Small stones may be passed out painlessly in the urine and may even go undetected.
- However, it is fairly common for a stone to block part of the urinary system

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Kidney Stones (renal calculi)

Signs & Symptoms

- Persistent ache in the lower back / groin / side of the abdomen
- Periods of intense pain in back/side of the abdomen, lasting minutes or hours (renal colic)
- Feeling restless and unable to lie still
- Need to urinate more often
- Pain when urinating
- Blood in the urine

Nausea and/or vomiting

Can lead to UTI

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Acute Kidney Injury (AKI)

AKI is a sudden episode of kidney failure or kidney damage that happens within a few hours or days.

It can range from minor loss of kidney function to complete kidney failure.



AKI causes a build-up of waste products in the blood making it hard for the kidneys to maintain the right balance of fluid within the body.

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Acute Kidney Injury (AKI)

It's essential that AKI is detected early and treated promptly.

Without quick treatment, abnormal levels of salts and chemicals can build up in the body, which affects the ability of other organs to work properly.

If the kidneys shut down completely, this may require temporary support from a dialysis machine, or lead to death.

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Acute Kidney Injury (AKI)

Signs & Symptoms

- Nausea and/or vomiting
- Less than 100ml urine excretion in 24 hours
- Diarrhoea
- Dehydration
- High blood pressure
- Confusion
- Drowsiness
- Abdominal pain / back ache



Most cases of AKI are caused by reduced blood flow to the kidneys.

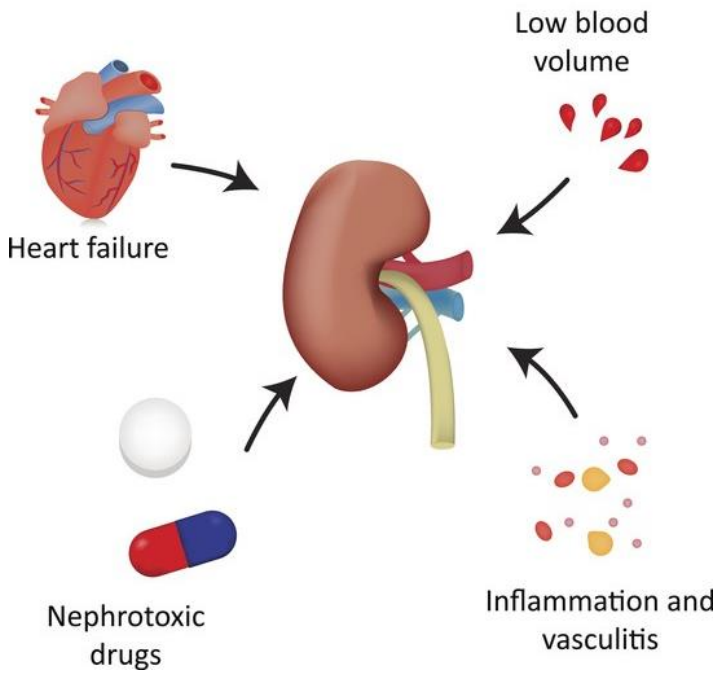
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Acute Kidney Injury (AKI)

Reduced blood flow could be caused by:

- Low blood volume
- The heart pumping out less blood than normal
- Problems with the blood vessels
- Certain medicines



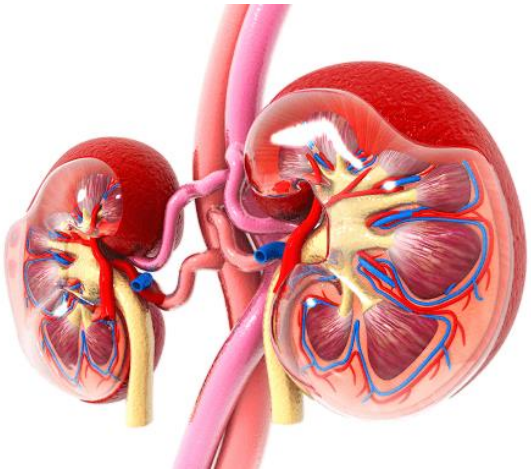
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Acute Kidney Injury (AKI)

AKI can be caused by a problem with the kidney itself or as a result of a blockage affecting the drainage of the kidneys, such as:

- An enlarged prostate
- A tumour in the pelvis, such as an ovarian or bladder tumour
- Kidney stones



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Acute Kidney Injury (AKI)

Treatment of AKI depends on the cause of the illness and how severe it is. Treatment may include:

- Increasing intake of water and other fluids if dehydrated
- Antibiotics if an infection is present
- Stop taking certain medicines (at least until the problem is resolved)
- A urinary catheter, to drain the bladder if there's a blockage

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Acute Kidney Injury (AKI)

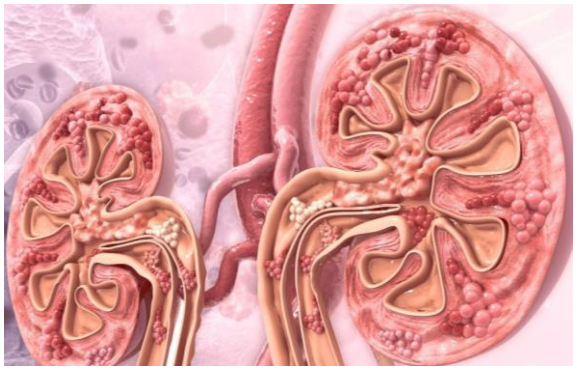
Serious complications of AKI are:

- High levels of potassium in the blood – in severe cases, this can lead to muscle weakness, paralysis and heart rhythm problems
- Fluid in the lungs (pulmonary oedema)
- Acidic blood (metabolic acidosis) – which can cause nausea, vomiting, drowsiness and breathlessness

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Chronic Renal Failure



Also known as chronic kidney disease (CKD), is a long-term condition where the kidneys don't function as well as they should.

A common condition often associated with getting older & can affect anyone.

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Chronic Renal Failure

There are usually no symptoms of kidney disease in the early stages. At a more advanced stage, signs & symptoms can include:

- Tiredness
- Swollen ankles, feet or hands
- Shortness of breath
- Feeling sick
- Blood in urine
- Nausea/vomiting
- Hypertension

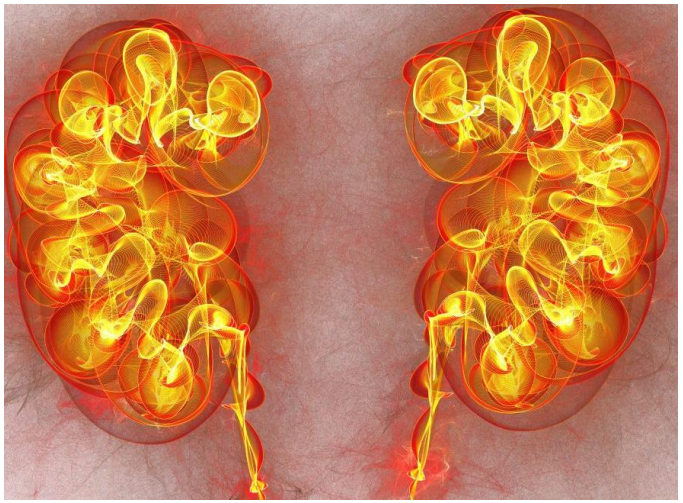


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Chronic Renal Failure

CKD is usually caused by other conditions that put a strain on the kidneys, including:



- High blood pressure
- Diabetes
- High cholesterol
- Kidney infections
- Glomerulonephritis
- Polycystic kidney disease
- Blockages in the flow of urine
- Long-term, regular use of certain medicines

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Chronic Renal Failure

Treatment of CKD depends on the cause of the illness and how severe it is. Treatment may include:

- Lifestyle changes
- Medicine to control associated problems
- Dialysis
- Kidney transplant



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Chronic Renal Failure

- CKD can range from a mild condition with no or few symptoms, to a very serious condition where the kidneys stop working, sometimes called kidney failure.
- Most people with CKD will be able to control their condition with medicine and regular check-ups.
- People with CKD, even if it's mild are an increased risk of developing other serious problems, such as cardiovascular disease.

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Management

- *Oxygen – as per oxygen JRCALC guidelines*
- Administer pain relief as required, consider assistance
- Constant observations
- Consider sepsis and treat as per guidelines
- Undertake a 12 lead – *elderly & those with cardiac risks factors & upper abdominal pain*
- Transfer to further care

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Questions



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